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Supplementary Information for

Measuring Dynamic Media Bias

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This PDF file includes:

Supplementary text

Supporting Information Text

Extended Materials and Methods

The full data set used for this study, in addition to code that can be used to build the dynamic media bias model will be publicly available upon publication via Harvard Dataverse. We also include additional specifications of the measure, e.g., that do not include Donald Trump, that do not include politicians, in an extended appendix available there.

Data Collection and Verification. Our primary data source is the Stanford Cable TV News Analyzer database. With permission from the Stanford team, we scraped the amount of time each of the 1,695 people their facial recognition algorithm detects were on screen across CNN, FoxNews, and MSNBC from January 1, 2010 through February 24, 2021. We then dropped anchors of shows, based on the Stanford team's documentation (https://tvnews.stanford.edu/data/people), leaving us with 1,225 unique names and their associated screen time, and over 1.5 million channel-person-day observations.

Next, to get the ideology of each individual in the Stanford data, we merged it with the Database on Ideology, Money in Politics, and Elections. This was a multi-step process to match against all 29 million observations in the original DIME dataset. First, we assigned a gender to each individual in the Stanford data and joined the names into the DIME data based on an exact match of first name, last name and gender. We also created a separate joined dataset by only last name and gender. Before describing the process through which we verified these matches, it is important to note that, in general, each unique contributor is assigned one ID in the DIME data and their ideological score does not vary within that ID. However, upon manual inspection of the matches we discovered that for many individuals there were clearly multiple IDs over time. If this is the case, we use the time-varying ideology as their score – in other words, if there is an observation for a person in 2012 and 2016, under different IDs, their ideology will vary over time. For others with only one match (which should be the case for most), their ideology does not vary within the sample.

Since our primary measure of media bias is created by weighting a channel-day's DIME-CF score based on how much time an individual is on the screen, it was important to verify the matches to the DIME dataset. We had a team of five research assistants manually investigate each of the 20,000 possible matches and code as a mismatch, possible match, or certain match based on information provided in the DIME data (such as occupation and location) and information available on the internet about each person. The authors then manually inspected a random sample of the certain matches and all uncertain matches to verify the check done by the research assistants. Finally, we manually examined all unmatched names to alleviate concerns around false negatives based on variations in name spelling. This resulted in a final dataset of 704 individuals matched to DIME scores (excluding anchors which were not matched), or just over 1 million channel-person-day observations.

For our analyses and descriptive figures, we aggregate up these weighted scores per channel-day and then take the weekly average per channel. Because our measure of media slant relies heavily on the DIME estimation process, interrogating the validity of this measure requires demonstrating that such media bias is "visible" to the audience; in other words, we need to show whether the talk scripts by a speaker with a high CF score, when read with plain eyes, are deemed more conservative than a speaker with a low CF score. We employed two research assistants to read each transcript, acquired from LexisNexis, from broadcasts across each channel in June 2018. Each speaker within the transcript was anonymized to remove sources of bias if the research assistants were familiar an individual and used their priors, instead of the spoken words, to assess slant. We also noted if the speaker was an anchor specifically by calling them "anchor." Next, we asked each coder to go through each speaker-segment (i.e., each un-interrupted speech by an anchor/guest) and code as -1 for liberal, 0 for neutral, or 1 as conservative. Coders independently went through each transcript, with inter-coder reliability scores (weighted Cohen's Kappa) of 0.664 (CNN), 0.528 (MSNBC), and 0.662 (Fox). Then, for each show, we averaged the slant scores to aggregate up to the show-day level. Finally, to correspond with our measure, we create a weekly ideology score for each show and compare it to the show's screen time-based measure, with a correlation between 0.65 and 0.72 depending on the sample.

We suggest this is qualitatively comparable to text-scaling based methods such as those employed in Gentzkow and Shapiro (2010) as well as Martin and McCrain (2019), but allows comparability to the granular nature of the screen time-based measure. Automated text-scaling approaches only use a fraction of the actual text in the scaling exercise and do not take into account context which, as a result, would produce very high variance in a day-to-day measure (or anything finer grained).

Finally, we compared our measure to that created by Padgett et al. (1) tallied statements made by each member of congress between the 109th and 112th congress (2005-2013). We created an analogous weighted ideology measure (number of statements x DW-Nominate score) for each member. Aggregating our measures to the congressional terms that overlap with that data (111th and 112th congress), we find strong correlations between the two datasets ($r_{cnn} = .76$,$r_{foxnews} = .74$,$r_{msnbc} = .85$.)

56 1. J Padgett, JL Dunaway, JP Darr, As seen on tv? how gatekeeping makes the us house seem more extreme. *J. Commun.* **69**, 696–719 (2019).